WORKED SOLUTION

STEP I 1992, Question 9

Pure • Differential Equations

STEP I 1992, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q9

The key idea

Everything follows from one equation: the rate of change of the *volume* of water equals (rate in) minus (rate out). By similar triangles the volume is a simple function of the depth x, so the chain rule turns this into a differential equation for x. In the first part the equation separates neatly; in the second, rather than solving it, we read off the behaviour from the sign of $\mathrm{d}x/\mathrm{d}t$ and from the depth at which it vanishes.

Solution

Emptying the cone

When the water is x cm deep it fills a smaller cone, similar to the whole filter, of height x and radius r. By similar triangles $r/x = a/H$, so $r = ax/H$, and the volume of water is

$$V = \tfrac{1}{3}\pi r^2 x = \frac{\pi a^2 x^3}{3H^2}.$$

By the chain rule,

$$\frac{\mathrm{d}V}{\mathrm{d}t} = \frac{\mathrm{d}V}{\mathrm{d}x}\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{\pi a^2 x^2}{H^2}\frac{\mathrm{d}x}{\mathrm{d}t}.$$

Water only leaves, at rate Ax, so $\mathrm{d}V/\mathrm{d}t = -Ax$. Hence

$$\frac{\pi a^2 x^2}{H^2}\frac{\mathrm{d}x}{\mathrm{d}t} = -Ax \quad\Longrightarrow\quad x\frac{\mathrm{d}x}{\mathrm{d}t} = -\frac{AH^2}{\pi a^2},$$

where we divided by x, which is allowed while there is still water in the cone ($x > 0$). The left-hand side is $\frac{\mathrm{d}}{\mathrm{d}t}(\frac{1}{2}x^2)$, so $\frac{1}{2}x^2$ decreases at the *constant* rate $AH^2/(\pi a^2)$. It starts at $\frac{1}{2}h^2$ and the cone is empty when it reaches 0, so the time taken is

$$T = \frac{\frac{1}{2}h^2}{AH^2/(\pi a^2)} = \frac{\pi a^2 h^2}{2AH^2} \text{ seconds},$$

as required. (Equivalently, separate variables: $\int_h^0 x\,\mathrm{d}x = -\frac{AH^2}{\pi a^2}\int_0^T \mathrm{d}t$.)

Pouring water in as well

Now $\mathrm{d}V/\mathrm{d}t = B - Ax$, so

$$\frac{\pi a^2 x^2}{H^2}\frac{\mathrm{d}x}{\mathrm{d}t} = B - Ax, \qquad \text{that is} \qquad \frac{\mathrm{d}x}{\mathrm{d}t} = \frac{H^2(B - Ax)}{\pi a^2 x^2}.$$

Since $x > 0$, the factor $H^2/(\pi a^2 x^2)$ is positive, so $\mathrm{d}x/\mathrm{d}t$ has the sign of $B - Ax$. Write $c = B/A$. Then

$$\frac{\mathrm{d}x}{\mathrm{d}t} > 0 \text{ when } x < c, \qquad \frac{\mathrm{d}x}{\mathrm{d}t} = 0 \text{ when } x = c, \qquad \frac{\mathrm{d}x}{\mathrm{d}t} < 0 \text{ when } x > c.$$

So $x = c$ is an equilibrium depth, where inflow exactly balances outflow, and the level always moves *towards* c. This is only physically possible inside the filter if $c \le H$, so the cases are decided by comparing c with h and with H.

Ask yourself If the level is moving towards c, does it ever actually get there, or overshoot?

It can't overshoot: to pass c it would first have to equal c, and there $\mathrm{d}x/\mathrm{d}t = 0$. In fact it never reaches c in finite time. Treating t as a function of x, the time to go from h to a depth X between h and c is

$$t = \int_h^X \frac{\pi a^2 x^2}{H^2(B - Ax)}\,\mathrm{d}x = \int_h^X \frac{\pi a^2 x^2}{AH^2(c - x)}\,\mathrm{d}x.$$

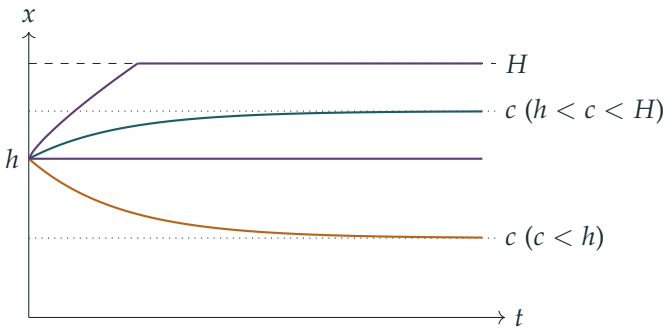
Near $x = c$ the integrand behaves like $\dfrac{\pi a^2 c^2}{AH^2}\cdot\dfrac{1}{c - x}$, whose integral is a multiple of $-\ln(c - x)$; this tends to infinity as $X \to c$. So the level approaches c ever more slowly and only reaches it "after infinite time". The same argument works from either side.

The cases are therefore:

- $B < Ah$ (so $c < h$). Initially $\mathrm{d}x/\mathrm{d}t < 0$: the level falls, slowing down, and tends to $c = B/A$ from above without reaching it. In particular, unlike in the first part, the cone never empties (as $c > 0$).
- $B = Ah$ (so $c = h$). Then $\mathrm{d}x/\mathrm{d}t = 0$ from the start: water flows in as fast as it drains, and the level stays at h.
- $Ah < B < AH$ (so $h < c < H$). The level rises, slowing down, and tends to B/A from below. It stays below H, so the filter never overflows.
- $B = AH$ (so $c = H$). The level rises towards the brim and tends to H, but by the argument above never quite reaches it, so (in this model) it never overflows.
- $B > AH$ (so $c > H$). Now $B - Ax \ge B - AH > 0$ for all $x \le H$, and $x^2 \le H^2$, so

  $$\frac{\mathrm{d}x}{\mathrm{d}t} \ge \frac{H^2(B - AH)}{\pi a^2 H^2} = \frac{B - AH}{\pi a^2} > 0.$$

  The rate of rise is bounded below by a positive constant, so the level reaches the brim in a finite time, at most $\pi a^2(H - h)/(B - AH)$ seconds. After that the filter overflows: the level stays at H and the excess $B - AH$ $\mathrm{cm}^3\,\mathrm{s}^{-1}$ spills over the top.



The sketch shows typical behaviour: falling to c (amber), staying at h or overflowing after a finite time (plum), and rising to c (teal).

Checks

- Putting $B = 0$ gives $c = 0$: the level falls towards 0, matching the first part (there it *does* reach 0, because the outflow Ax is proportional to x while the volume goes like x^3, so $\mathrm{d}x/\mathrm{d}t = -AH^2/(\pi a^2 x)$ does not tend to 0).
- Numerically, with $a = 3$, $H = 8$, $A = 0.7$, $h = 5$, the formula gives $T \approx 7.889\,\mathrm{s}$, and stepping x^2 in time gives the same. Integrating the second equation numerically confirms each case; for instance with $B/A = 9 > H$ the level reaches H after about 41.5 s, while with $B/A = 7$ the time to get within ε of 7 grows by about 71 s each time ε is divided by 10, the logarithmic growth predicted above.

Common slips

- Using the cylinder volume $\pi a^2 x$ instead of the volume of the similar cone of height x.
- Claiming the level reaches B/A in a finite time, or that it overshoots and oscillates.
- Forgetting the brim: the equilibrium B/A is only reached if $B/A \leq H$; otherwise the filter overflows.

Beyond STEP

The second part is a first taste of the *qualitative* theory of differential equations: for $\mathrm{d}x/\mathrm{d}t = f(x)$ you can find the equilibria (zeros of f), decide their stability from the sign of f on either side, and use uniqueness of solutions to show they are approached but never crossed, all without solving anything. Boyce and DiPrima, *Elementary Differential Equations*, treats autonomous equations and stability in this spirit.

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